

PesaSwap

Product Guide, User Stories & Visual Walkthrough

Multi-Currency Payment Platform for African Commerce

QR Ordering | M-Pesa Payments | Staff Management | AI Intelligence

Version 3.0 | May 2026

github.com/chmunyas/merchantApp

This document covers every screen with user stories, acceptance criteria, and screenshots showing the end-to-end customer and merchant experience.

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Part 1: Customer Journey (Mobile)

The customer experience is designed for zero friction: scan a QR code at the table, browse the menu, order, pay, tip, and leave a review - all without downloading an app or creating an account. Every interaction happens on a single screen with a bottom navigation bar.

1.1 Table Landing - Scan QR or Enter Number

User Story:

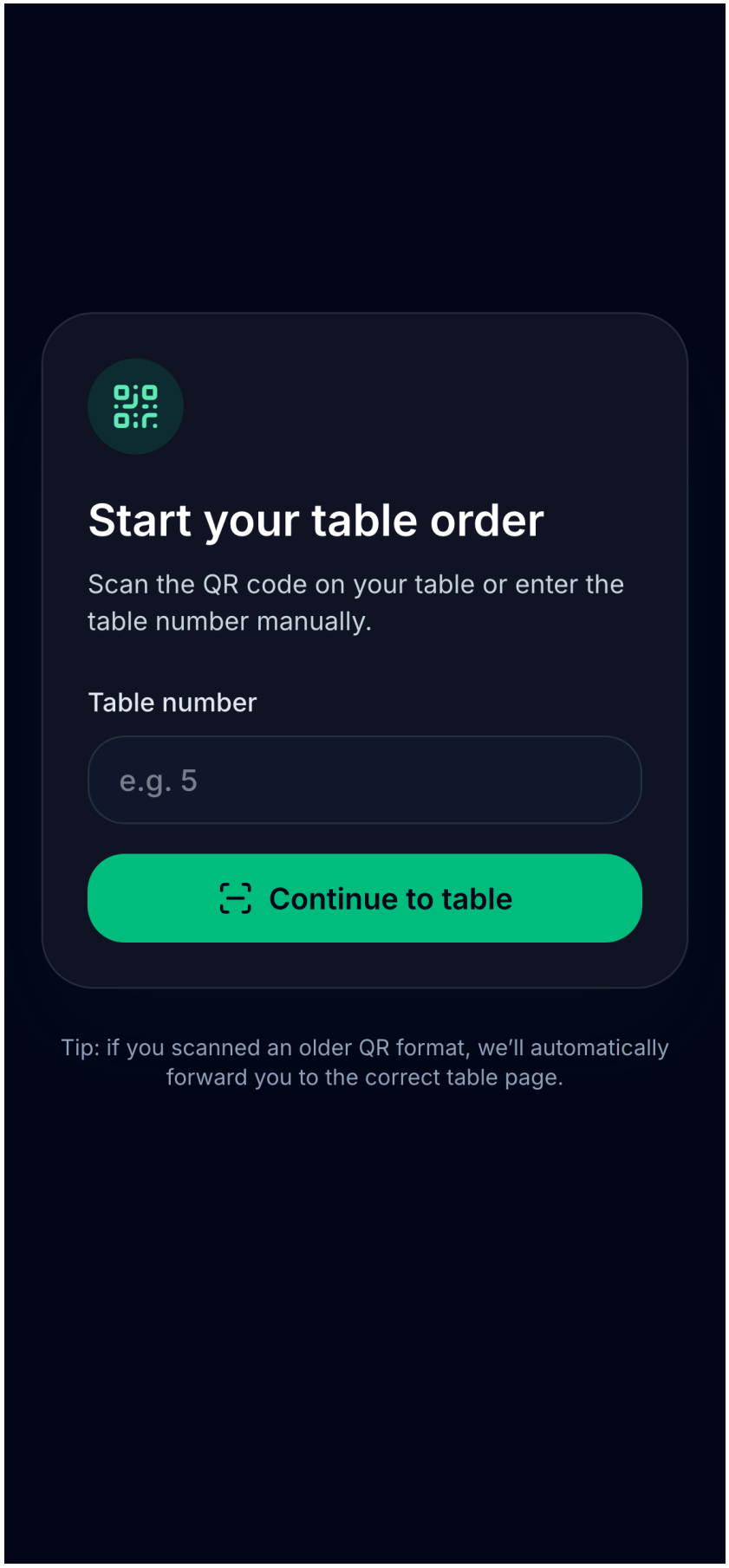
As a customer, I want to enter my table number or scan a QR code, so that I can start ordering immediately without confusion.

Acceptance Criteria:

[x] Landing page loads in <2 seconds

[x] Cl

Figure 1.1: Table entry landing page

The image shows a mobile app interface for starting a table order. It features a dark blue background with a central white rounded rectangle containing the main content. At the top left of this rectangle is a circular icon with a QR code. Below the icon is the heading 'Start your table order' in bold white text. Underneath the heading is a paragraph of white text: 'Scan the QR code on your table or enter the table number manually.' Below this is a label 'Table number' in white, followed by a white rounded rectangular input field containing the placeholder text 'e.g. 5'. Below the input field is a large, bright green rounded rectangular button with a white QR code icon and the text 'Continue to table' in white. At the bottom of the screen, below the white rectangle, is a line of white text: 'Tip: if you scanned an older QR format, we'll automatically forward you to the correct table page.'

1.2 Menu Browsing - Discover & Add Items

User Story:

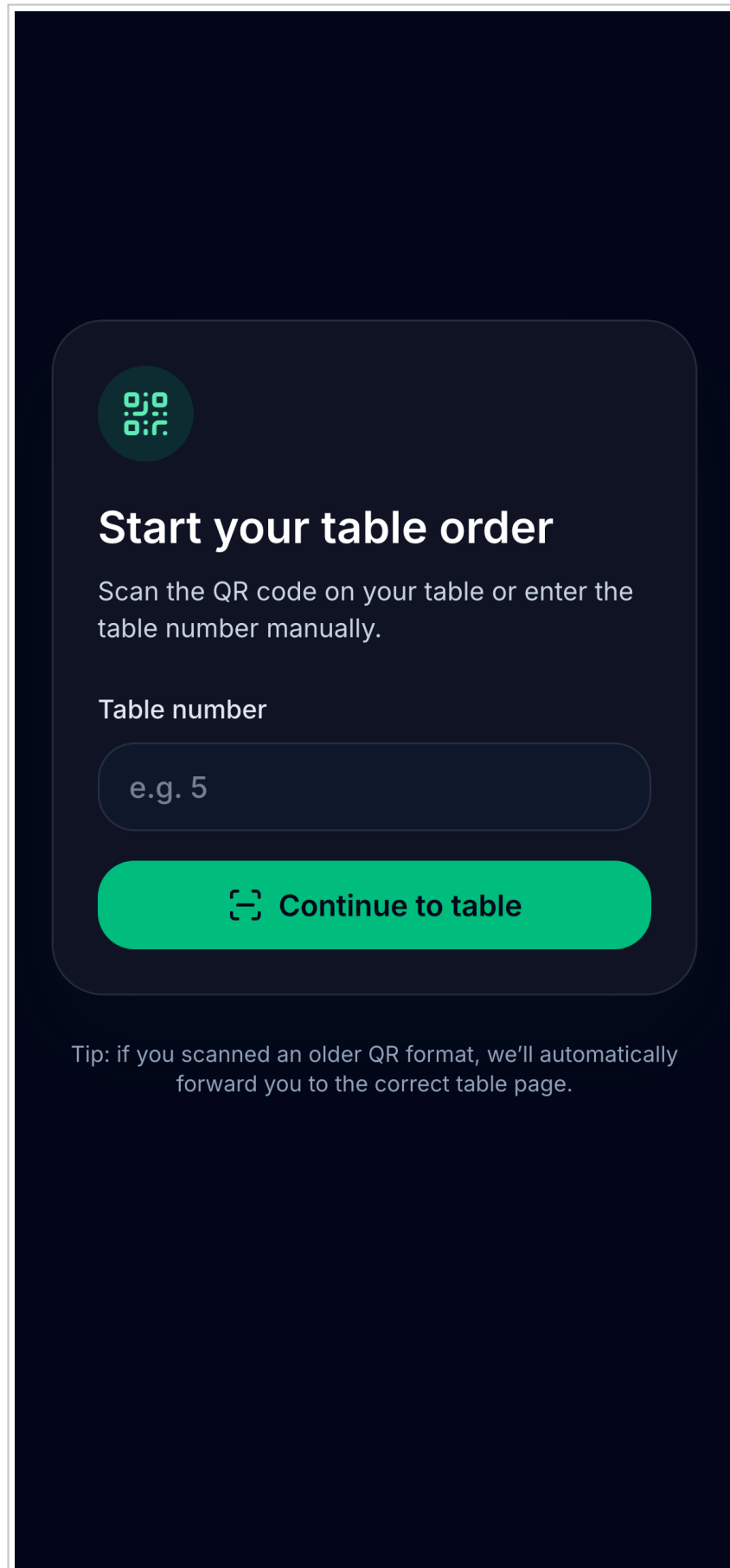
As a customer at Table 5, I want to browse the menu with photos and categories, so that I can quickly find what I want to eat and drink.

Acceptance Criteria:

[x] Menu loads within 2 seconds of entering table number

[x] Ite

Figure 1.2: Mobile menu browsing with categories and item cards



1.3 Adding Items - Customization & Modifiers

User Story:

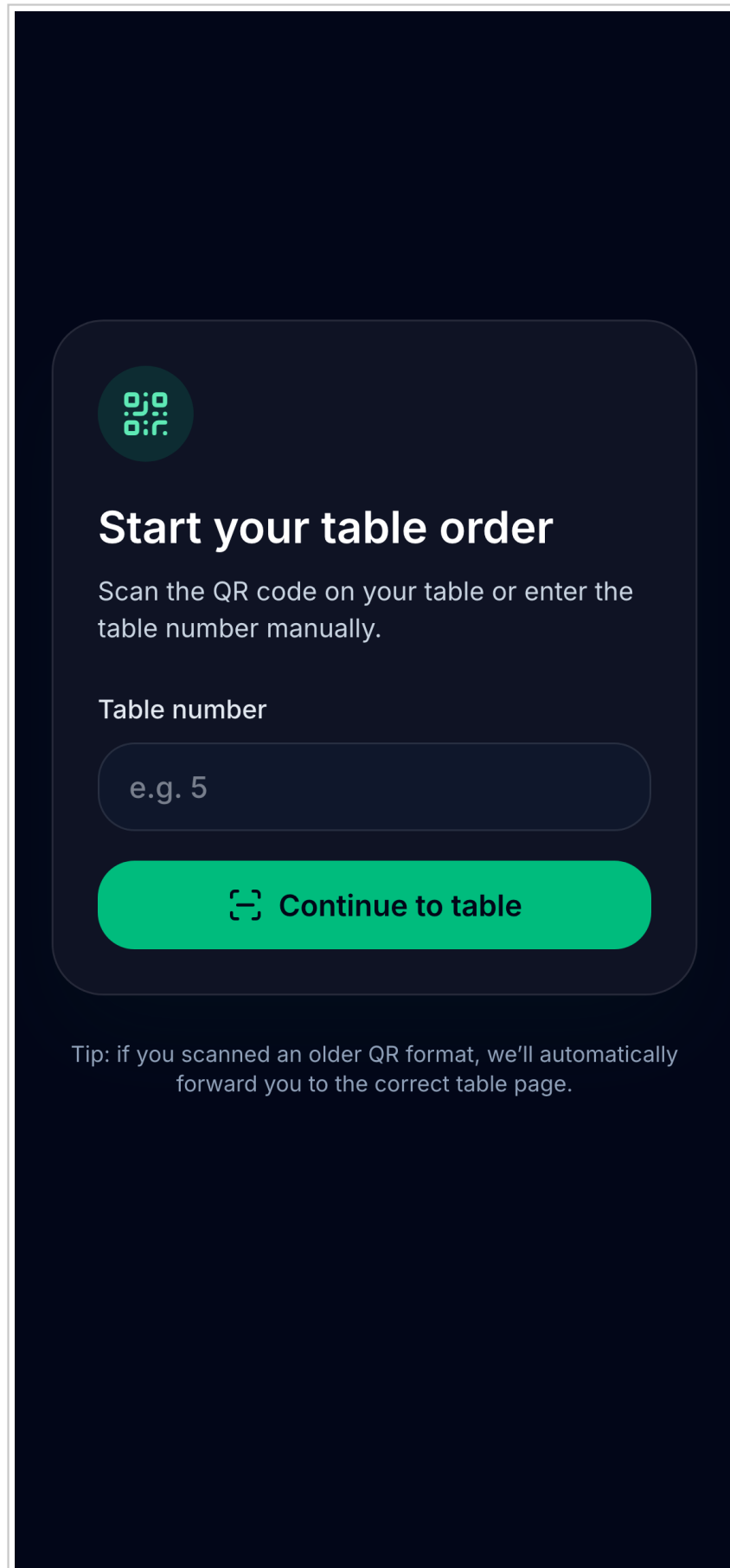
As a customer, I want to customize my order with size and extras, so that I get exactly what I want with clear pricing.

Acceptance Criteria:

[x] Tap "Add" shows modifier bottom sheet (if item has modifiers)

[x] M

Figure 1.3: Adding item with modifier selection



1.4 Cart Review - Adjust & Confirm Order

User Story:

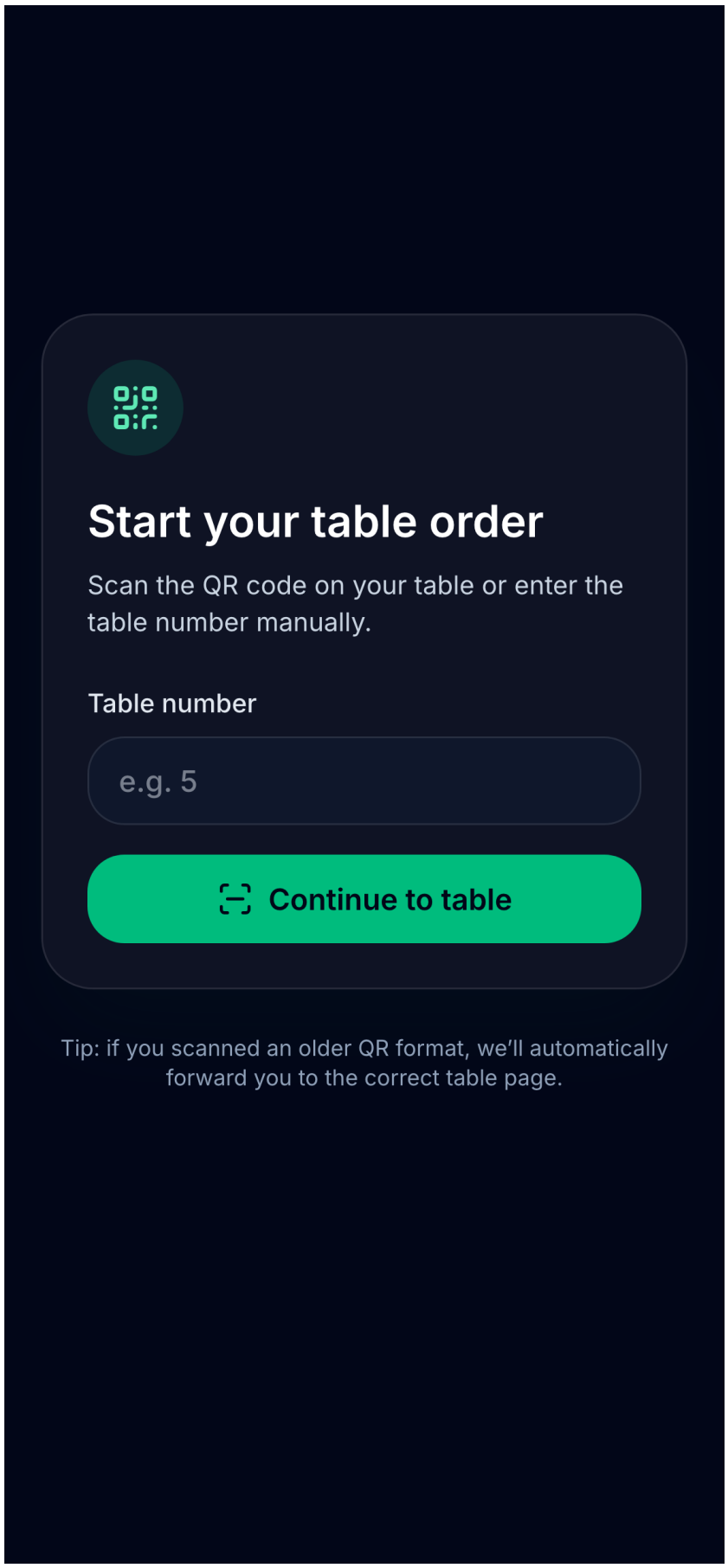
As a customer, I want to review my cart and adjust quantities before ordering, so that I do not accidentally order the wrong items or amounts.

Acceptance Criteria:

[x] Cart shows all items with name, modifiers chosen, qty, line total

[x] Pl

Figure 1.4: Cart review with quantity adjustment





Start your table order

Scan the QR code on your table or enter the table number manually.

Table number

e.g. 5

 Continue to table

Tip: if you scanned an older QR format, we'll automatically forward you to the correct table page.

1.5 Payment - Split, Tip & Pay (One Screen)

User Story:

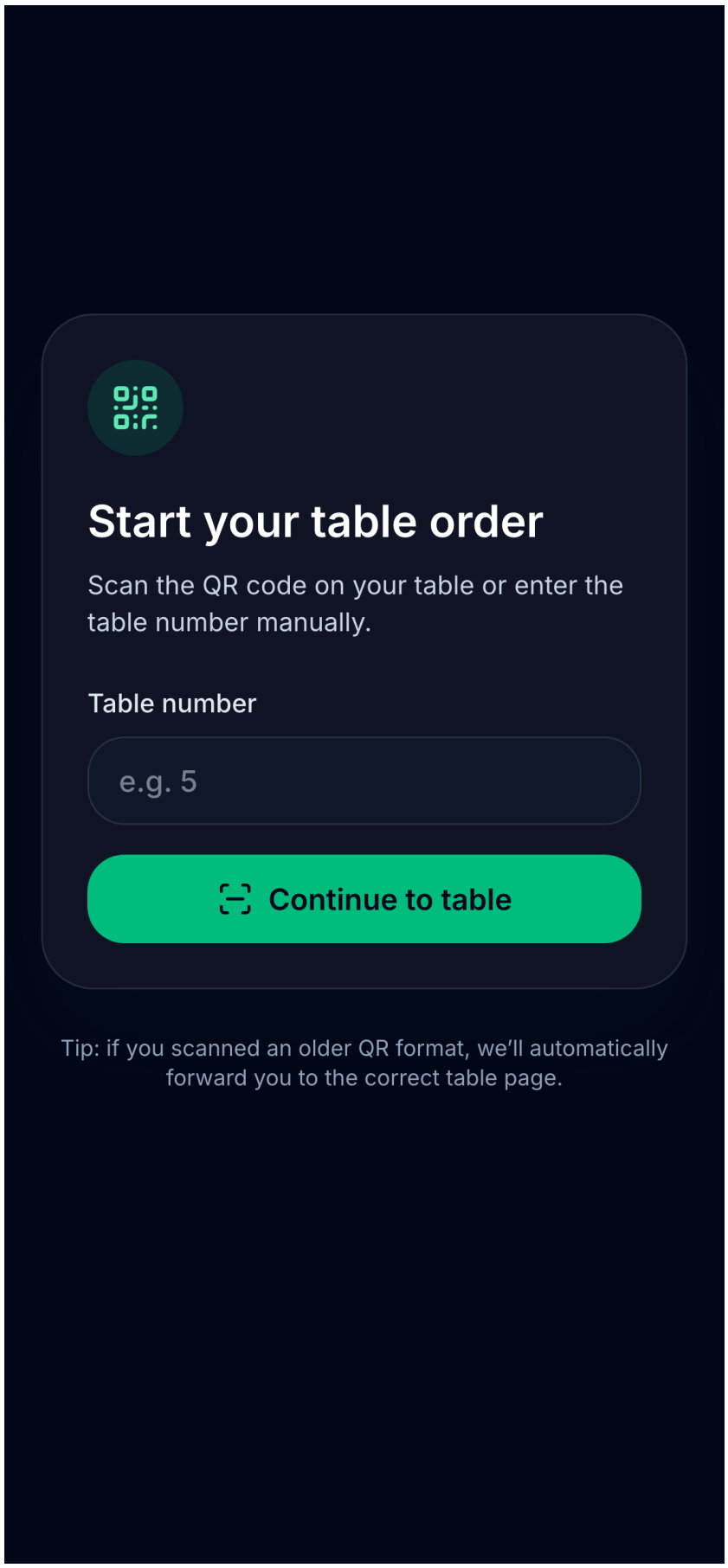
As a customer, I want to split the bill with friends, tip my server, and pay via M-Pesa all on one screen, so that I complete payment in under 10 seconds without navigating multiple pages.

Acceptance Criteria:

[x] Split options visible: Full | Equal Split | By Item | Custom Amount

[x] EO

Figure 1.5: One-screen payment with split, tip, and M-Pesa

A mobile app interface for starting a table order. It features a dark blue background with a central rounded rectangle containing a QR code icon, a title, instructions, a text input field, and a large green button. Below this rectangle is a tip paragraph.



Start your table order

Scan the QR code on your table or enter the table number manually.

Table number

e.g. 5

 Continue to table

Tip: if you scanned an older QR format, we'll automatically forward you to the correct table page.

1.6 Pre-Order Mode - Order Before Arriving

User Story:

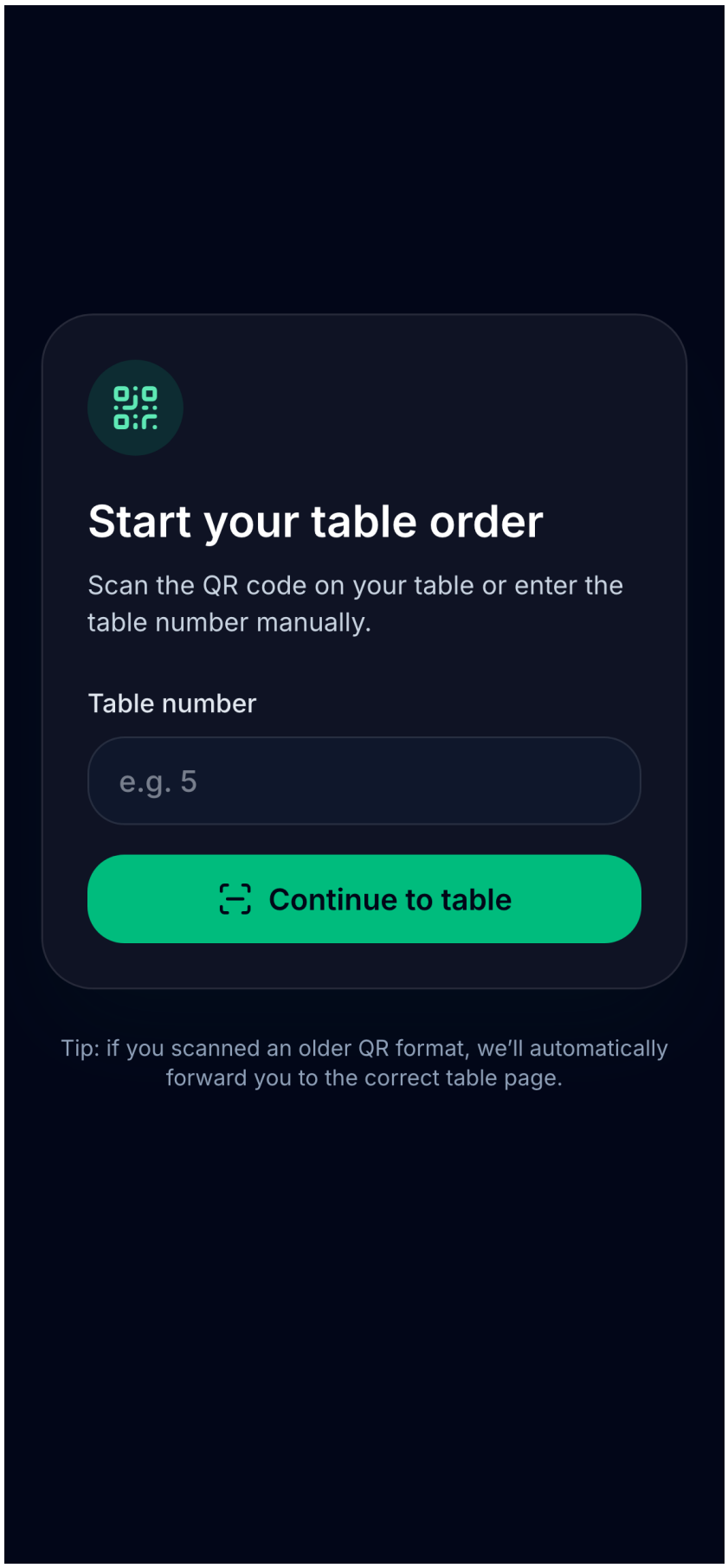
As a customer planning a dinner, I want to browse the menu and pre-order before I arrive, so that my food starts preparing when I get to the restaurant.

Acceptance Criteria:

[x] URL /table/5?preorder=true shows "Pre-Order" banner

[x] Fu

Figure 1.6: Pre-order mode with arrival time selection





Start your table order

Scan the QR code on your table or enter the table number manually.

Table number

e.g. 5

 Continue to table

Tip: if you scanned an older QR format, we'll automatically forward you to the correct table page.

Part 2: Merchant Dashboard (Desktop)

The merchant dashboard provides comprehensive business management: real-time revenue tracking, payment processing, invoicing, analytics, and customer review management. Designed for tablet/desktop use by restaurant owners and managers.

2.1 Landing Page & FX Rates

User Story:

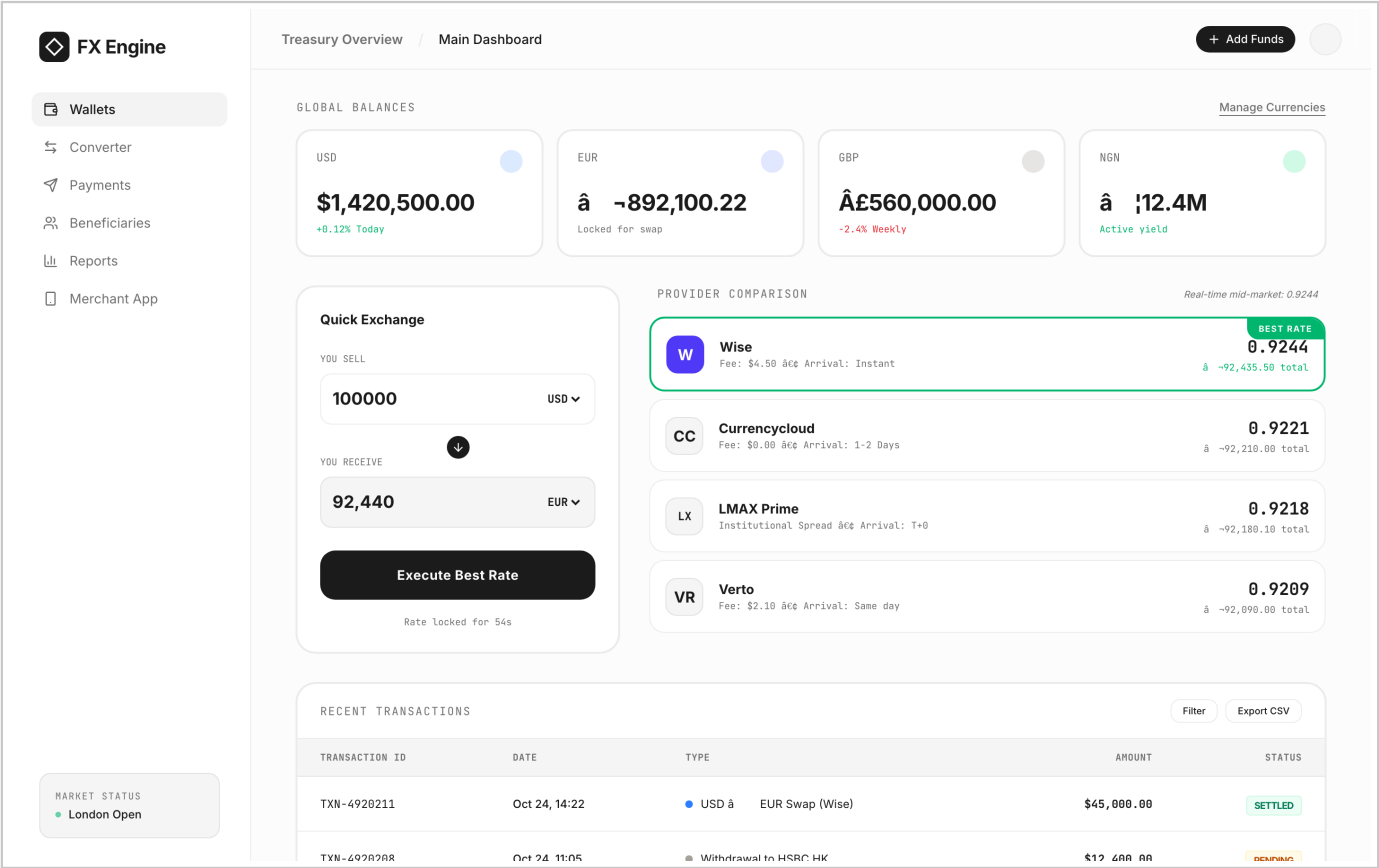
As a first-time visitor, I want to understand what PesaSwap does in under 5 seconds, so that I can decide whether to use it without reading documentation.

Acceptance Criteria:

[x] Clear value proposition headline visible above the fold

[x] Liv

Figure 2.1: Landing page with FX rates and value proposition



2.2 Merchant Dashboard Overview

User Story:

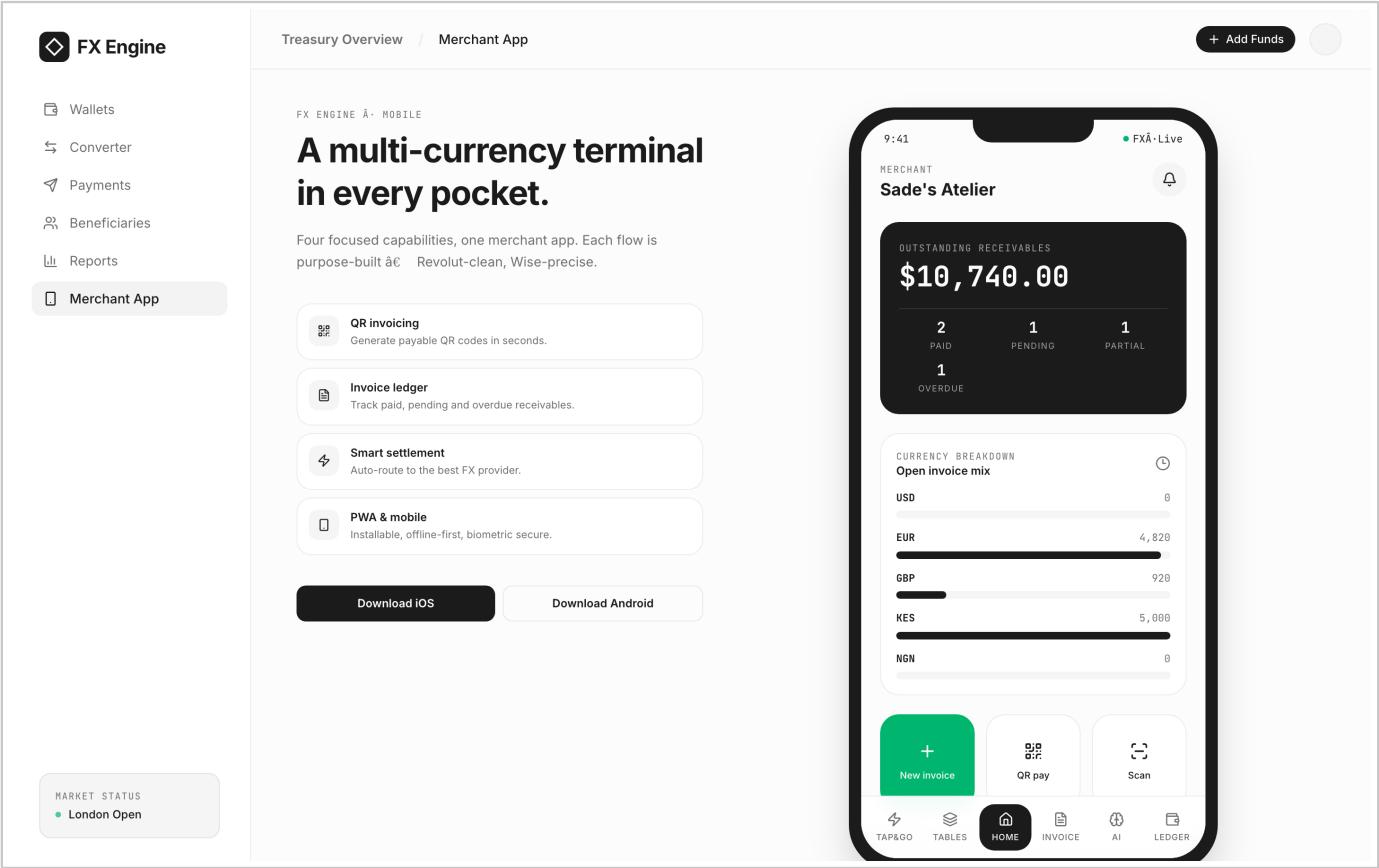
As a restaurant owner, I want to see my business health at a glance when I open the dashboard, so that I can make decisions without digging through reports.

Acceptance Criteria:

[x] KPI cards: Today's revenue, transaction count, avg ticket, active tables

[x] 7-

Figure 2.2: Merchant dashboard with KPIs and charts



2.3 Payments Management

User Story:

As a restaurant owner, I want to view all transactions and process refunds, so that I can reconcile my accounts and handle customer issues quickly.

Acceptance Criteria:

[x] Transaction list with: amount, method, status, date, customer

[x] Fil

Figure 2.3: Payment transaction management

Treasury Overview / Payments

+ Add Funds

RECENT TRANSACTIONS					Filter	Export CSV
TRANSACTION ID	DATE	TYPE		AMOUNT	STATUS	
TXN-4920211	Oct 24, 14:22	USD	EUR Swap (Wise)	\$45,000.00	SETTLED	
TXN-4920208	Oct 24, 11:05	Withdrawal to HSBC HK		\$12,400.00	PENDING	
TXN-4920201	Oct 23, 18:40	GBP	NGN Swap (Verto)	Â£8,200.00	SETTLED	
TXN-4920195	Oct 23, 09:12	Inbound EUR transfer		â -24,500.00	SETTLED	

2.4 Invoicing

User Story:

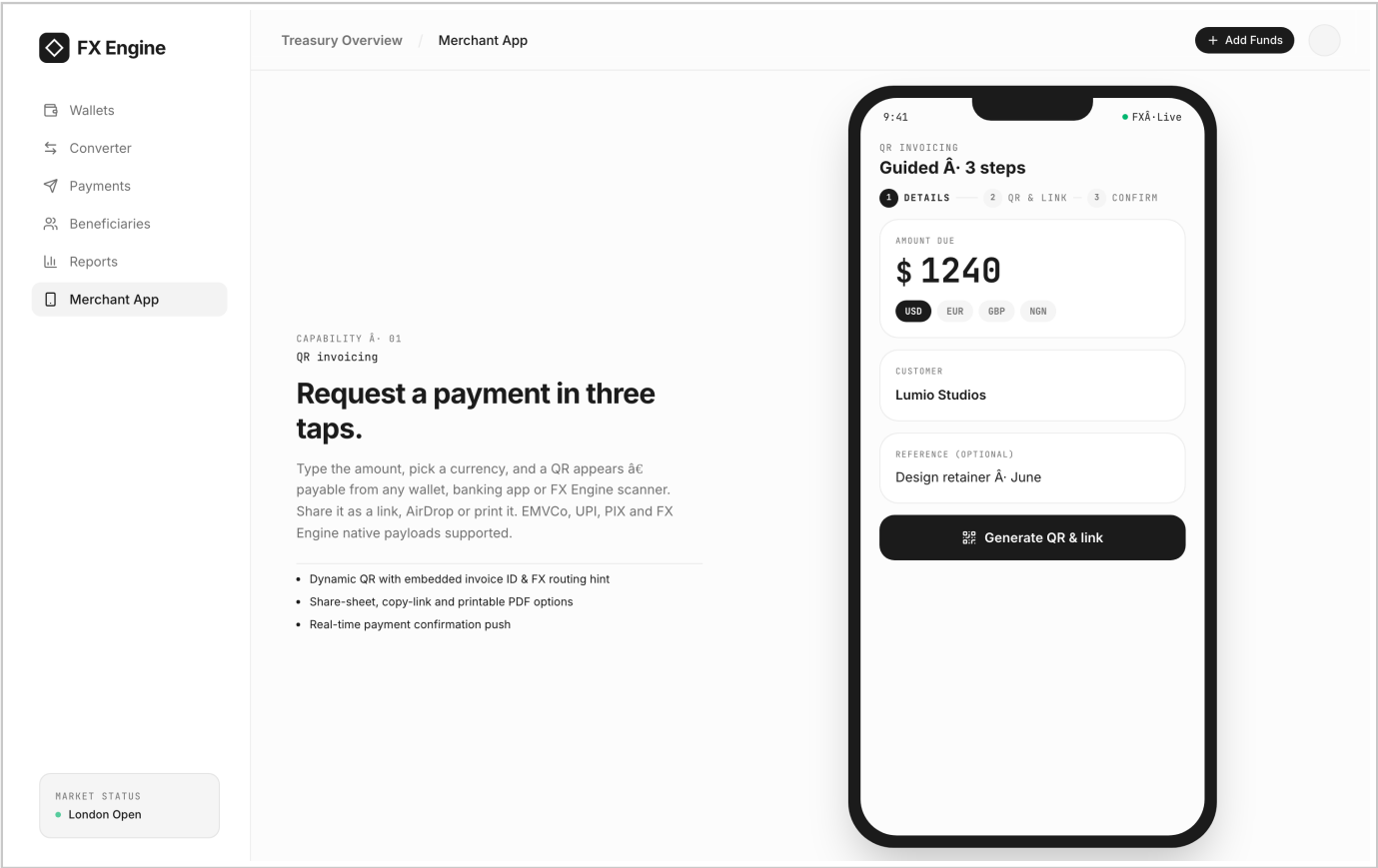
As a merchant, I want to create and send professional invoices in multiple currencies, so that I get paid faster with less follow-up.

Acceptance Criteria:

- [x] Create invoice with line items and tax calculations

[x] M

Figure 2.4: Invoice creation and management



2.5 Analytics & Insights

User Story:

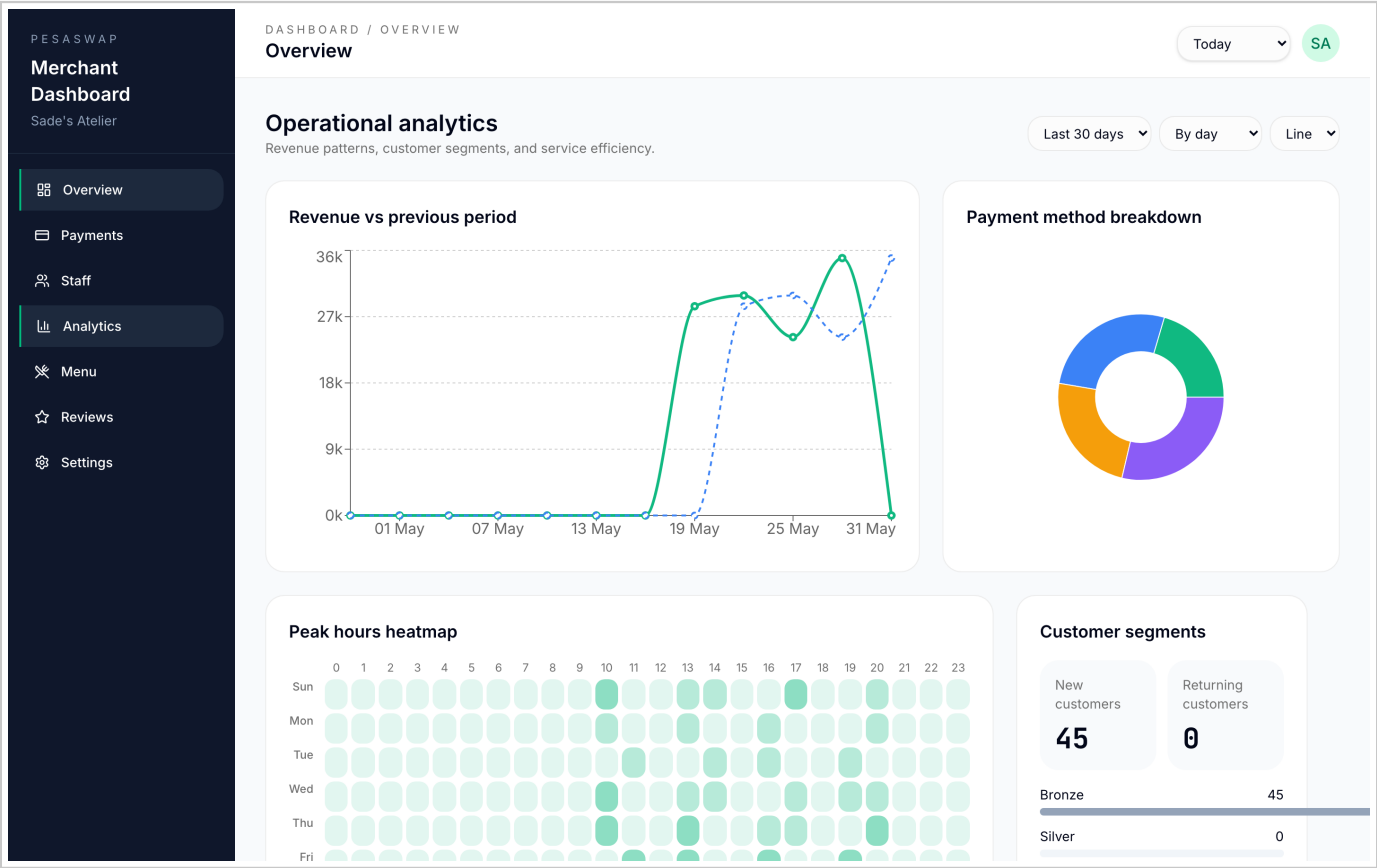
As a restaurant owner, I want to understand revenue trends and peak hours, so that I can optimize staffing and menu pricing.

Acceptance Criteria:

[x] Revenue charts by day/week/month

[x] Pe

Figure 2.5: Business analytics dashboard



2.6 Customer Reviews

User Story:

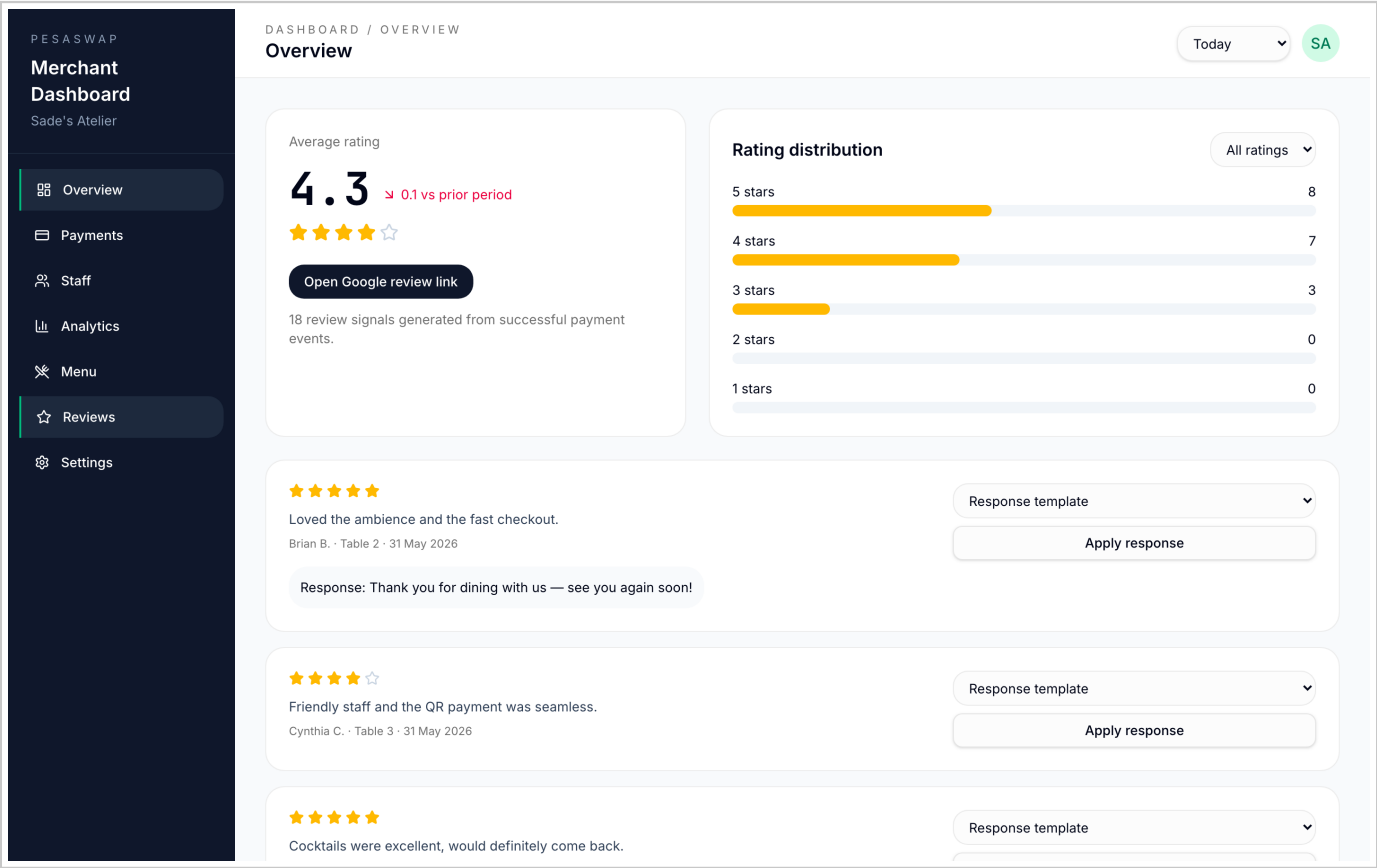
As a restaurant manager, I want to see all customer reviews in one place with ratings, so that I can identify service issues and celebrate wins with staff.

Acceptance Criteria:

[x] Review feed with star rating, comment, date, table number

[x] AV

Figure 2.6: Customer review management



Part 3: Menu Management

Full menu lifecycle management: create items with photos and modifiers, organize into multiple menus, assign to physical zones, and schedule time-based activation. All changes reflect immediately on customer-facing ordering.

3.1 Menu Items (Catalogue)

User Story:

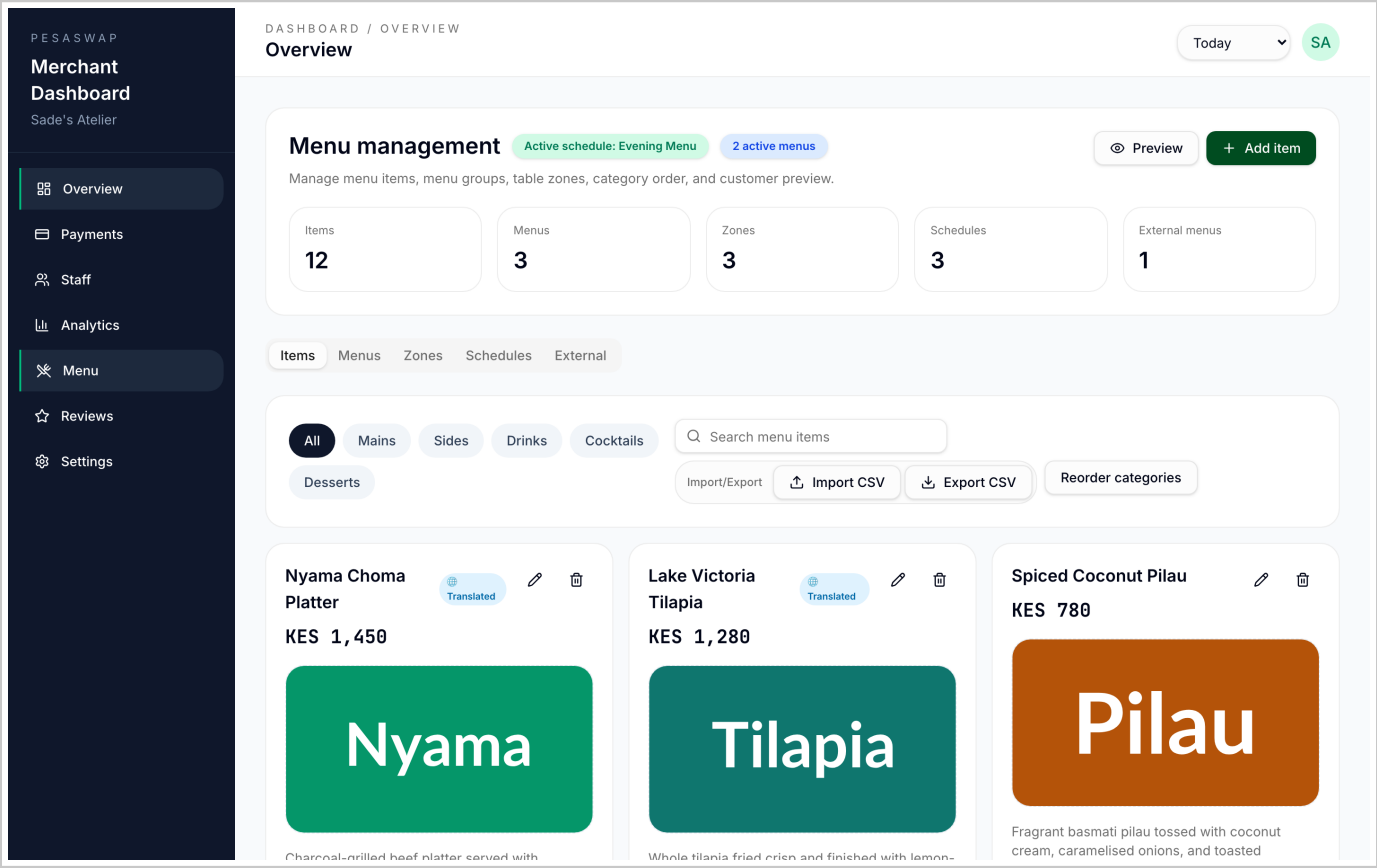
As a restaurant manager, I want to add, edit, and organize menu items with photos and modifiers, so that customers see an appetizing, accurate menu on their phones.

Acceptance Criteria:

[x] Add item: name, price, category, photo, description, dietary tags

[x] M

Figure 3.1: Menu item management with categories



3.2 Multiple Menus

User Story:

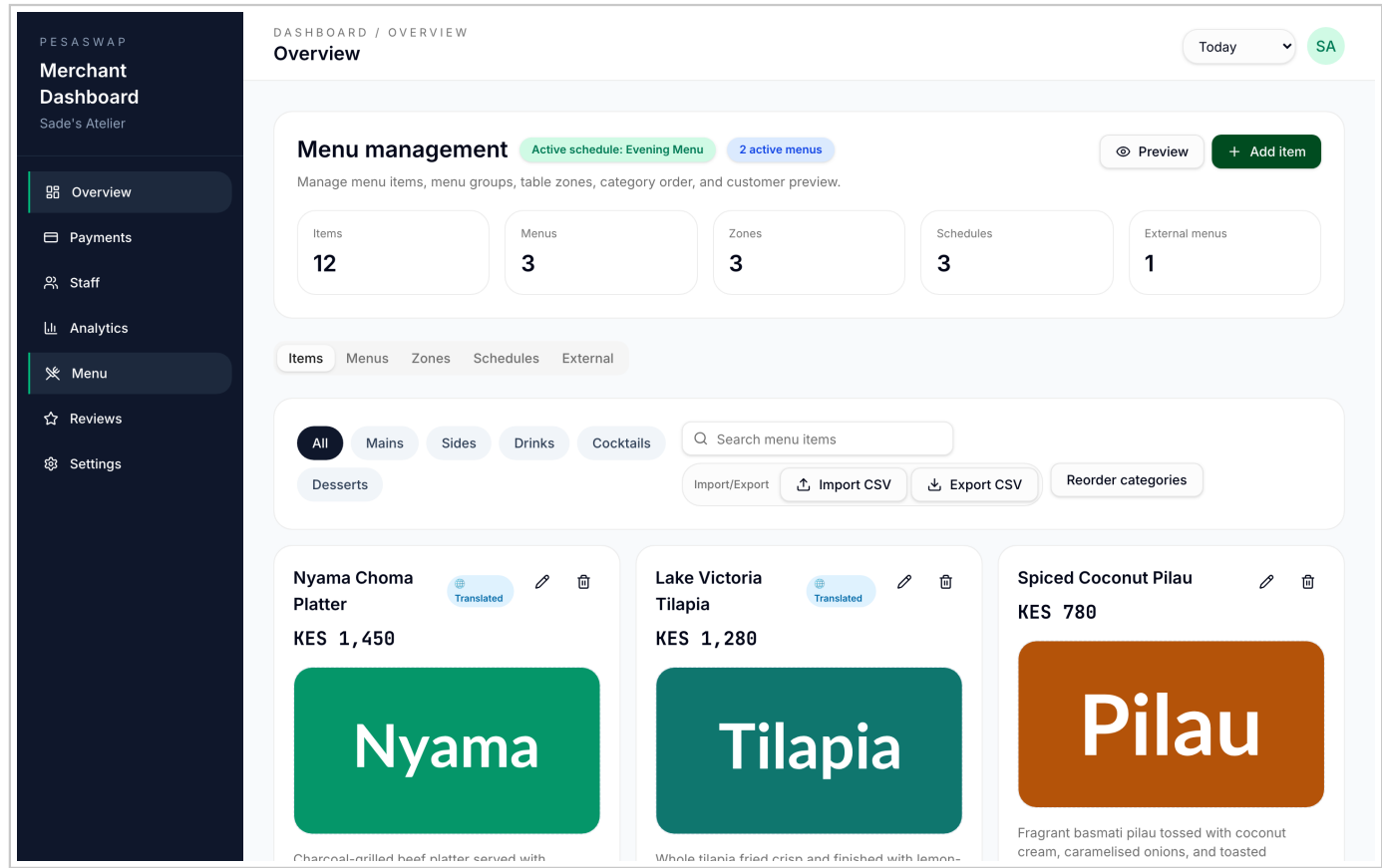
As a restaurant owner with breakfast/lunch/dinner service, I want to create separate menus that activate automatically by time, so that customers always see the right menu without staff intervention.

Acceptance Criteria:

[x] Create named menus (Breakfast, Lunch, Dinner, Happy Hour)

[x] As

Figure 3.2: Multiple menu management



3.3 Zones & Table Mapping

User Story:

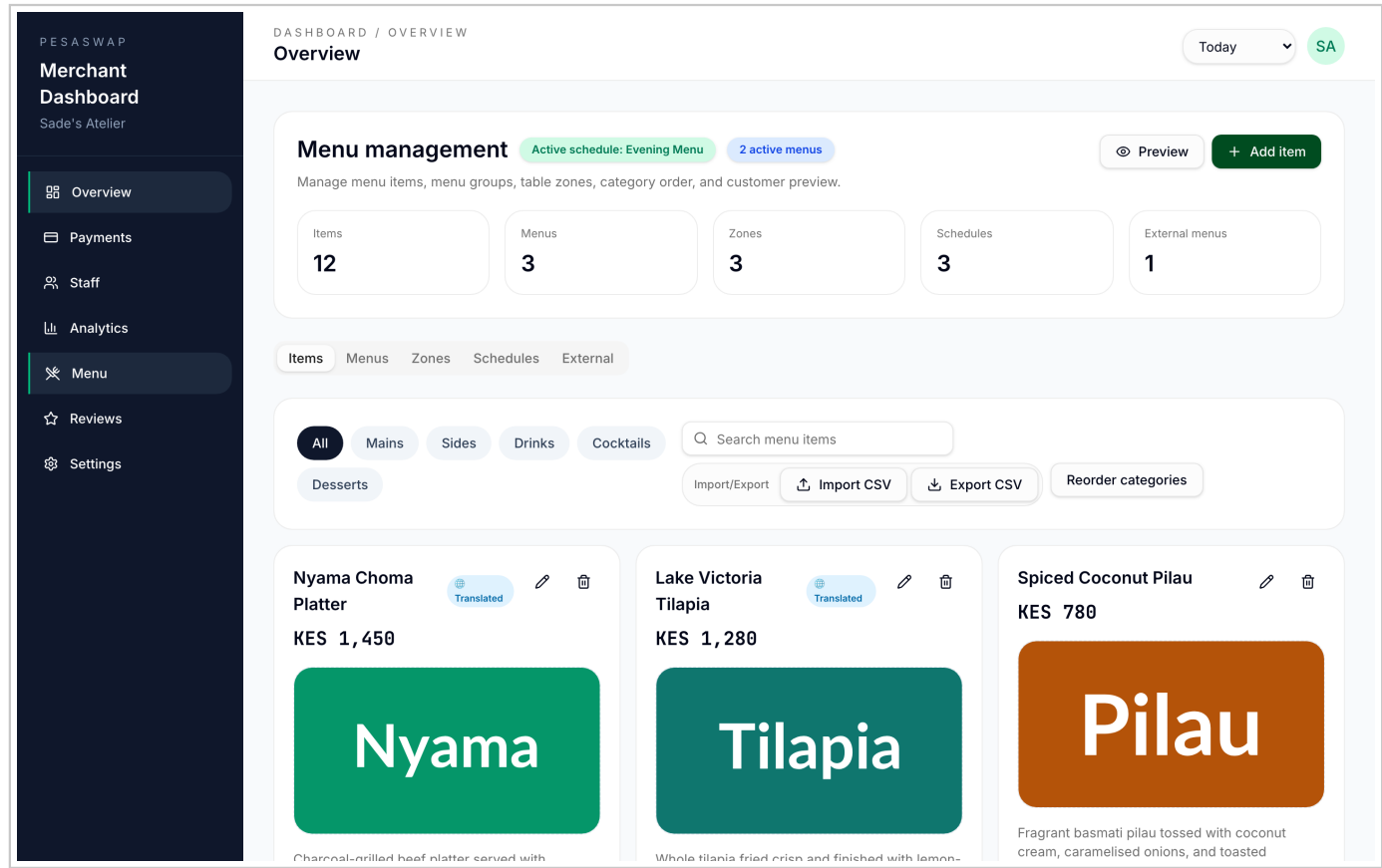
As a hotel manager with poolside and dining areas, I want to assign different menus to different physical areas, so that pool guests see drinks/snacks while dining guests see the full menu.

Acceptance Criteria:

[x] Create zones with name and table number ranges

[x] As

Figure 3.3: Zone configuration with table ranges



Part 4: Staff App (M-Pesa Enabled)

The Staff App replaces traditional bank-based tip distribution with instant M-Pesa payouts. Staff phone number IS their bank account. Includes performance gamification, shift scheduling, and AI coaching - all designed for the African market where mobile money is primary.

4.1 Team Management

User Story:

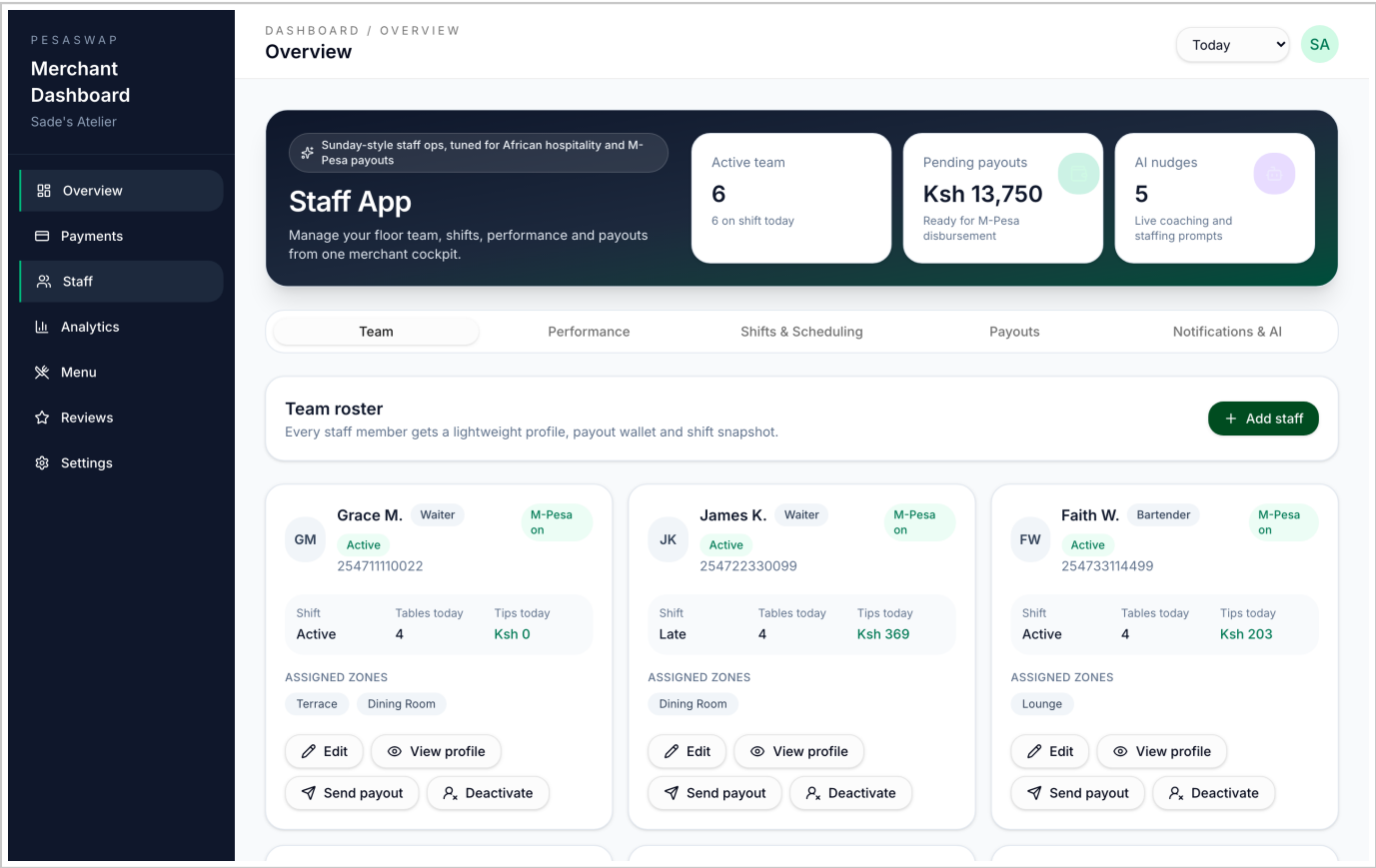
As a restaurant manager, I want to add and manage staff with their M-Pesa numbers and role assignments, so that tips and payouts go to the right people instantly.

Acceptance Criteria:

[x] Staff directory with photo/initials, name, role badge, phone, status

[x] AC

Figure 4.1: Staff team directory with role badges



4.2 Performance & Gamification

User Story:

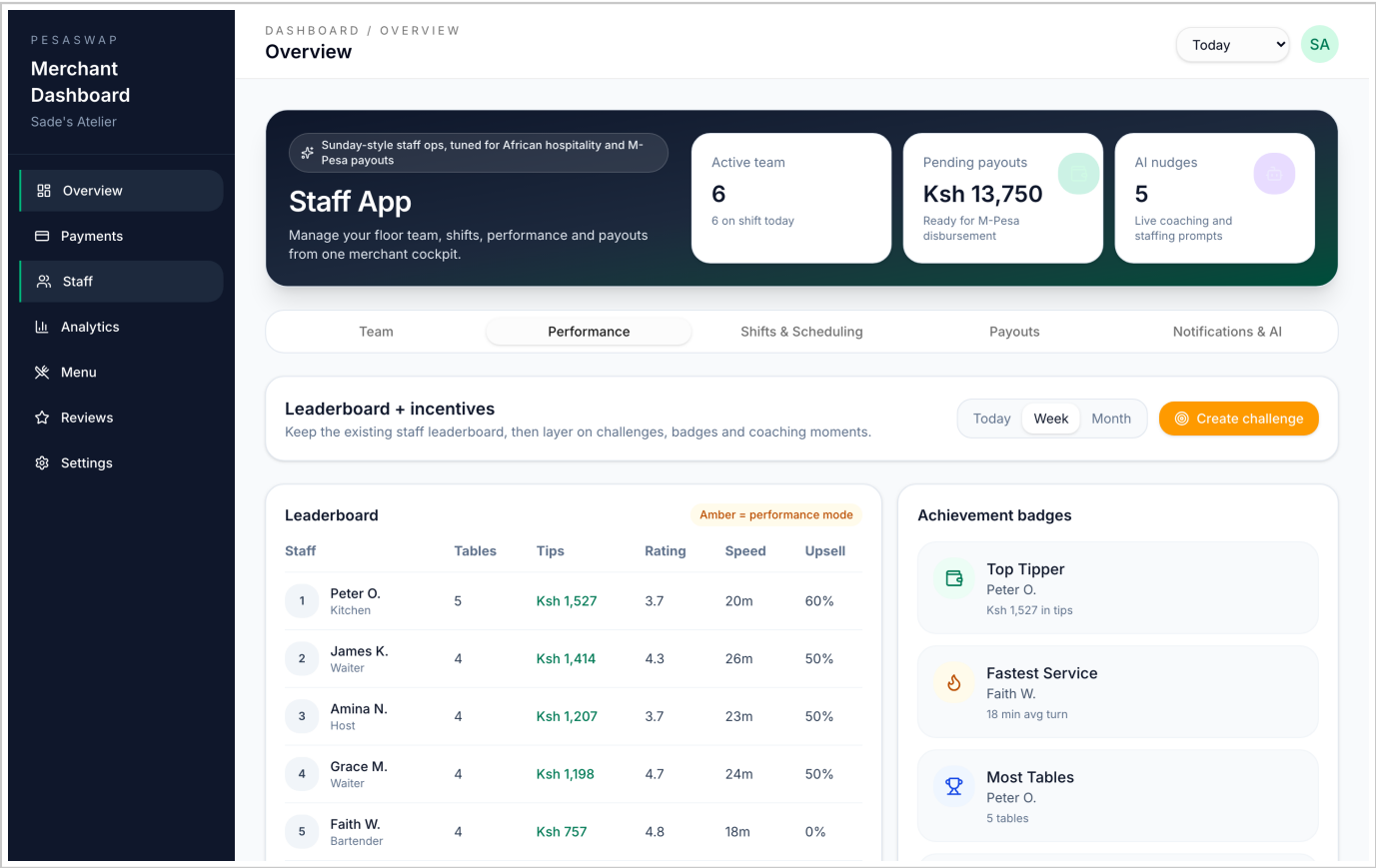
As a staff member, I want to see my ranking and compete in challenges for bonus money, so that I am motivated to provide better service and earn more.

Acceptance Criteria:

[x] Leaderboard ranked by: tips, tables served, rating, speed

[x] Pe

Figure 4.2: Performance leaderboard and challenges



4.3 Shift Scheduling

User Story:

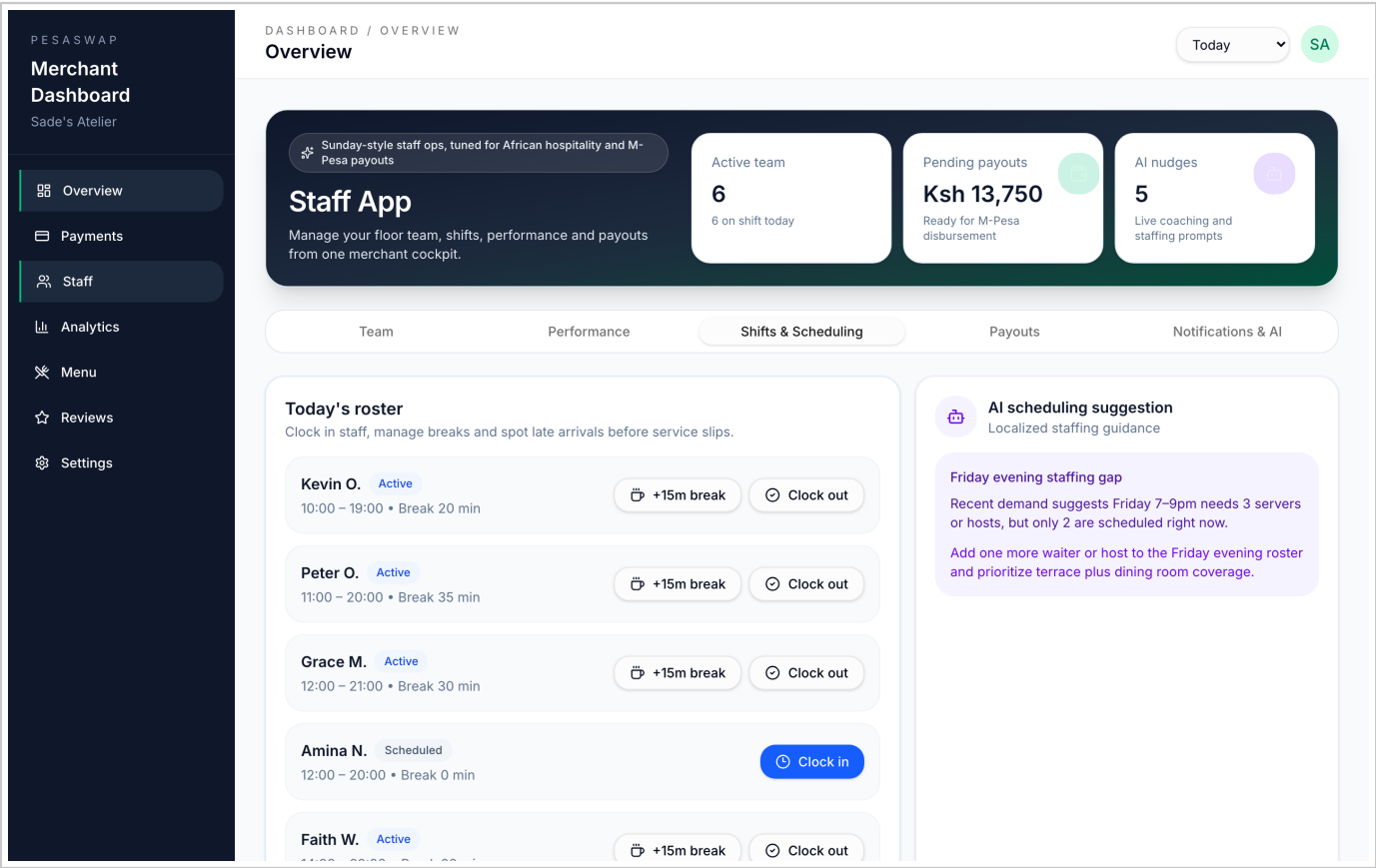
As a restaurant manager, I want to schedule staff shifts and track attendance, so that I always have the right number of people for each service period.

Acceptance Criteria:

[x] Weekly calendar view (Mon-Sun) with staff on Y-axis

[x] Cr

Figure 4.3: Weekly shift calendar with roster



4.4 M-Pesa Payouts

User Story:

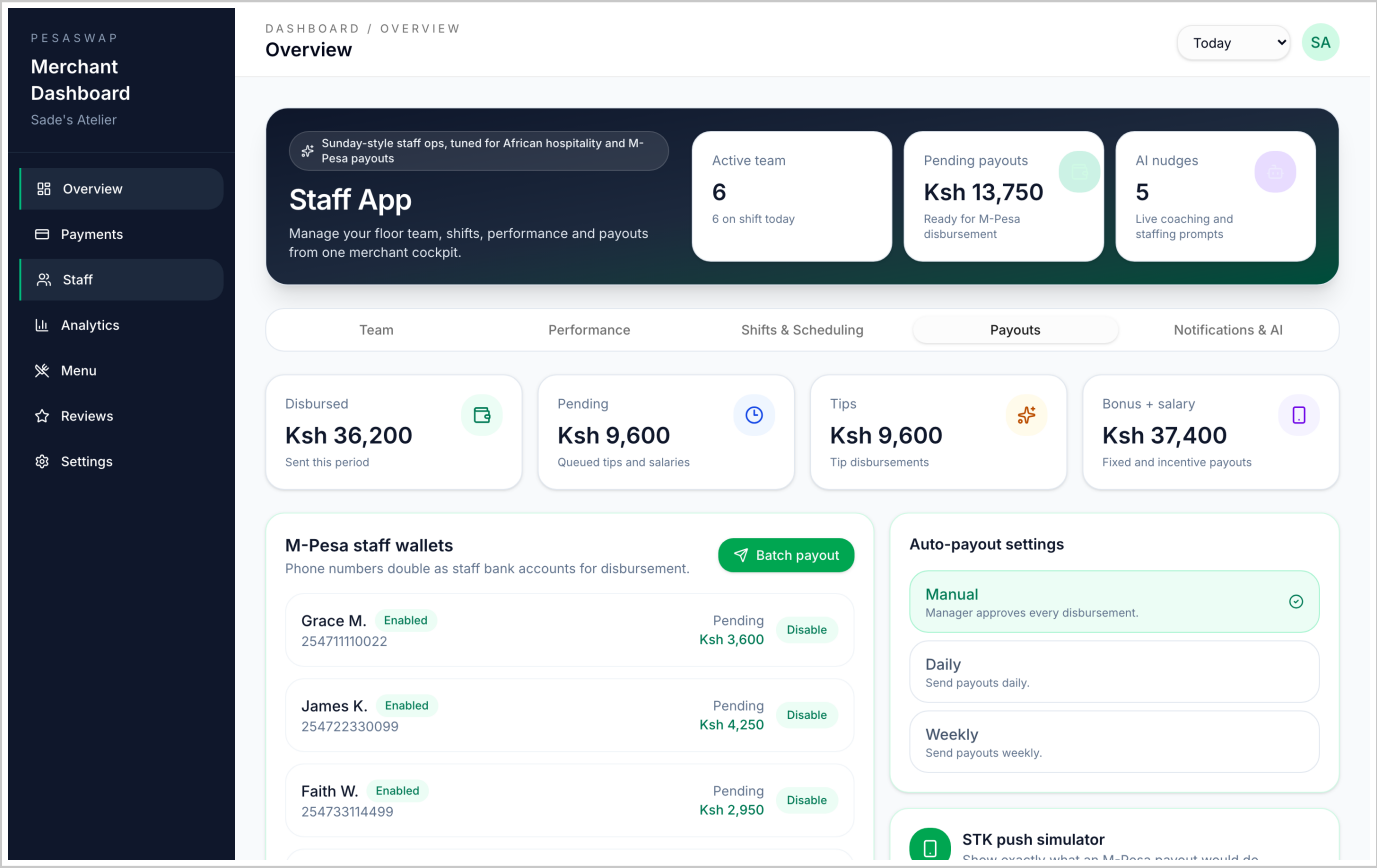
As a staff member, I want to receive my tips directly to my M-Pesa same day, so that I do not have to wait days for bank transfers or cash distribution.

Acceptance Criteria:

[x] Staff phone number = payout destination (no bank setup needed)

[x] In

Figure 4.4: M-Pesa payout management



4.5 Notifications & AI Insights

User Story:

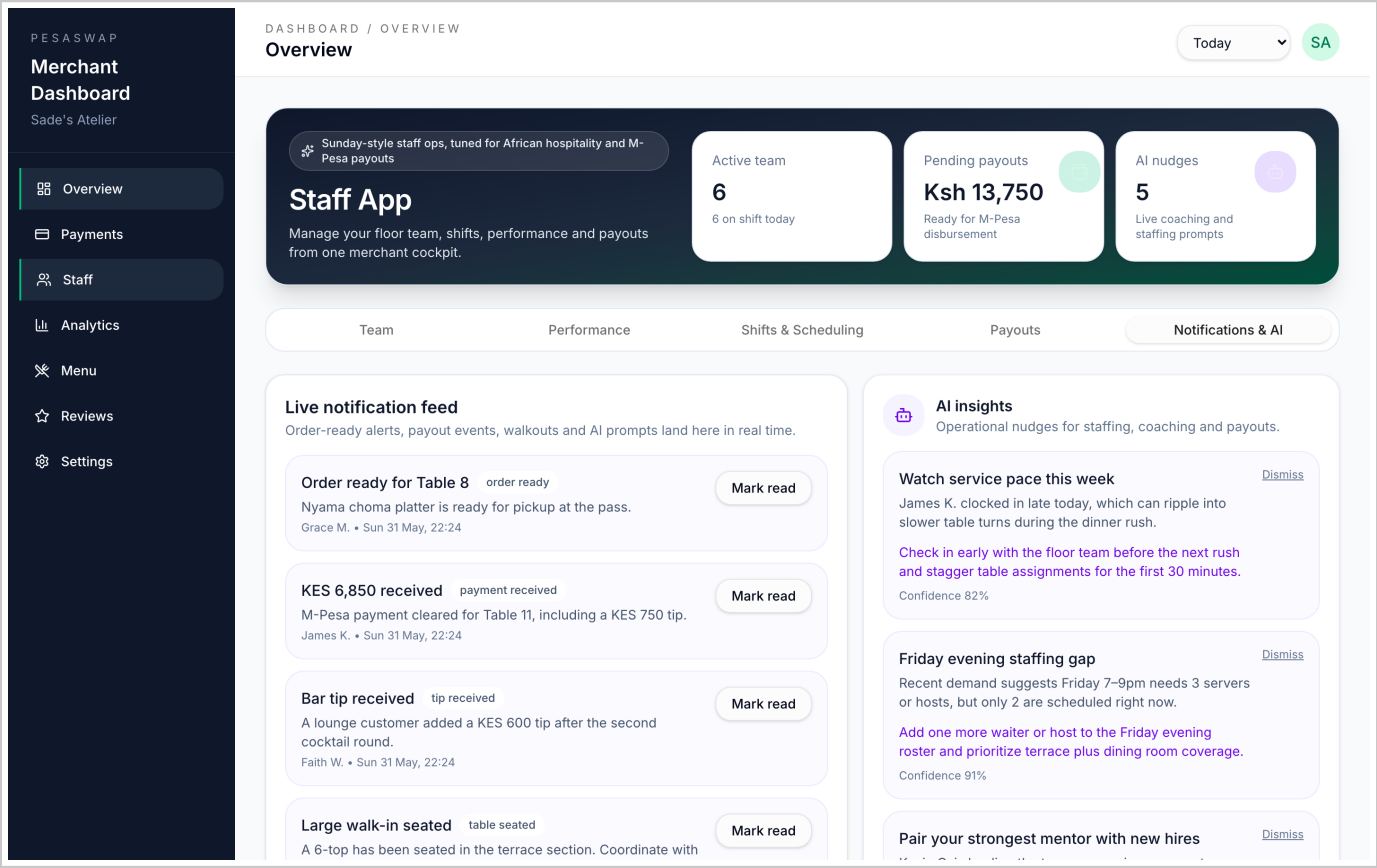
As a restaurant manager, I want to receive AI-powered insights about staffing and performance, so that I can proactively improve operations before problems escalate.

Acceptance Criteria:

[x] Real-time notification feed: order ready, payment, tip, walkout

[x] W

Figure 4.5: AI insights and notification feed



Appendix: Architecture & Getting Started

System Architecture

PesaSwap runs on edge infrastructure for global low-latency access:

Customer Layer	/table/\$tableId - Mobile ordering, payment, review
Merchant Layer	/dashboard - Management, staff, menu, analytics
Payment SDK	M-Pesa STK, Card, QR, one-tap - smart routing <100ms
FX Engine	Multi-provider rate comparison, best-rate settlement
Staff Layer	M-Pesa payouts, shifts, challenges, AI coaching
Intelligence	Context Graph ontology, decision traces, predictions
Infrastructure	Cloudflare Workers + R2 + D1 + KV (300+ PoPs)

Quick Start

- 1. git clone <https://github.com/chmunyas/merchantApp> && cd merchantApp

- 2. n

Key URLs

- /table/5 - Customer ordering (Table 5)

- /tab